



Sharing Session by Build Up Nepal and HRRP-Nepal

A green recovery - Affordable housing for vulnerable families and resilient livelihoods using Compressed Stabilized Earth Brick (CSEB) interlocking bricks in Nepal

2nd October 2020

A promotional banner for a sharing session. The background is dark blue with white and light blue geometric shapes. On the left, the text "SHARING SESSION" is in large, white, bold letters. Below it, a subtitle reads: "A green recovery - Affordable housing for vulnerable families and resilient livelihoods using CSEB interlocking bricks in Nepal." A dark blue button with white text says "REGISTER HERE". Below the button, it says "Organized by:" followed by logos for HRRP, Build Up Nepal, DCA, Practical Action, and Community Impact Nepal. On the right, a circular inset photo shows three people (two women and one man) working with a large orange brick-making machine outdoors. A vertical credit line on the right edge of the photo reads "Image Courtesy: Build Up Nepal".

PRESENTERS

- Mr. **Björn Söderberg**, Founder, Build up Nepal Engineering
- Mr. **Deepak Hamal**, Easy Bricks, Palung, Makwanpur
(President, Interlocking Bricks Association, Nepal)
- Mr. **Dadhiraj Pudasaini**, Indrasarovar Interlocking Bricks Udhyog, Makwanpur

FACILITATOR

- Mr. **Ruplal Aidi**, HRRP, National Coordination Officer.

ORGANIZERS

- Build Up Nepal Engineering.
- Housing Recovery and Reconstruction Platform (HRRP) - Nepal.

*Link to complete webinar recording [here](#).

A green recovery - Affordable housing for vulnerable families and resilient livelihoods using Compressed Stabilized Earth Brick (CSEB) interlocking bricks in Nepal

*Link to presentation [here](#).

*Link for sample test of CSEB in California [here](#).

BACKGROUND

Housing Recovery and Reconstruction Platform (HRRP) in collaboration with Build up Nepal Engineering, Dan Church Aid (DCA), Practical Action & Community Impact Nepal organized a Webinar Session on 02 October 2020 on the subject “A green recovery - Affordable housing for vulnerable families and resilient livelihoods using CSEB interlocking bricks in Nepal”. Mr. Björn Söderberg - Founder of Build up Nepal Engineering; Mr Deepak Hamal, the President of Interlocking Bricks Association of Nepal and the proprietor of Easy Bricks, Palung, Makwanpur; and Mr Dadhiraj Pudasaini, Indrasarovar Interlocking Bricks Udhyog presented papers on CSEB and discussed on the comparative advantages and CSEB, its benefit and support in low cost housing reconstruction along with nature conservation by reducing CO2 emission including entrepreneurship skill development at the local level. Fired bricks are responsible for 37% of Nepal’s CO2 emissions from combustion (World bank 2020). Can eco-friendly CSEB interlocking bricks replace fired bricks and boost a green recovery in Nepal? This is a subject matter for discussion and we need to debate on how we can prevent CO2 emissions. Most importantly, the event was provided as a platform to discuss how we can ensure that vulnerable families with poor financial capacity are not left behind in the reconstruction.

Following the presentation from the presenters and interactive discussion, Dr. Chandra Bahadur Shrestha, Executive Member of NRA provided valuable suggestions to the entrepreneurs, who presented on the manufacturing of CSEB and recommended liaising with the Department of Cottage and Small Industry for specification and quality assurance on the CSEB. Likewise, Mr. Shyam Kishor Singh, Project Director of CLPIU-Building, recommended to collaborate with local governments to increase acceptance on the use of CSEB and for further dissemination of such technology at local level.

INTRODUCTION

Most poor families in Nepal aspire to live in a brick house. Can CSEB interlocking bricks make this dream affordable? Build up Nepal, DCA and Practical Action & Community Impact Nepal shares their experience of introducing CSEB in more than 300 communities in Nepal,

building 5000 houses together with partners and local entrepreneurs.

By developing a network of local entrepreneurs, it is possible to build houses at low cost across multiple locations, making it an effective method to reach vulnerable families living in challenging locations. Each enterprise creates 5-10 jobs within the community, generating resilient livelihood long after the project is completed.

KEY POINTS HIGHLIGHTED DURING THE PRESENTATION

- Advantages of CSEB interlocking bricks in comparison to fired bricks
- Social acceptance and technical challenges
- Cost comparison and how to build a house in less than NPR 400,000 for those who were left behind in the reconstruction process and the most vulnerable.
- Community driven construction or working through local entrepreneurs - effectivity of each model
- Creating resilient livelihood, long-term development reducing CO2 emissions - the NABIN project
- Building for most vulnerable in Leave no-one behind

SHARING AND PRESENTATION BY MR. BJORN SODERBERG, BUILD UP NEPAL

After the 2015 earthquake, people started building houses following the guidelines of NRA. Many of the buildings reconstructed after the earthquake are of brick which has been used to construct houses long before. People are most familiar with this construction material but have very little knowledge on Compressed Stabilized Earth Brick (CSEB) interlocking brick. In the context of Nepal, the rural house construction is a challenging one where difficulty of material transportation and lack of trained and experienced workers can lead to earthquake prone houses/buildings. Immediately after the earthquake, Build Up Nepal was focusing on different local technologies like the use of bamboo, earth-bags and stone masonry with GI wire. Most of the households were happy after the construction but they did not replicate them at all.

CSEB is a very simple technology and affordable to everyone. This technology has been introduced to 300+ communities and constructed more than 5000 houses in different locations of Nepal through entrepreneurship. Vulnerable families with very poor financial capacity should not be left behind in the reconstruction process.

Build Up Nepal is working in close consortium with DCA, Practical action, Community Impact

Nepal. It's not just for the five years of the earthquake but there are still other households who will build their houses in future. The most vulnerable, scattered location settlements are very difficult to reach. A scalable solution that can be self-sustained and work as well because most of other projects are also at the end of their budgets. Most of the households desired to construct houses with brick and cement. With this, the cost of construction at remote areas will go high and is not affordable at rural communities. To achieve low cost construction, one must achieve the social acceptance with social aspirations. Even a poor family would prefer to construct a brick and cement house.

CHALLENGES

- Fired bricks produce 37% of Co2 emissions in Nepal
- Making brick houses affordable and Eco-friendly with low cost.

Fired brick production has even caused 28% of air pollution in Kathmandu valley. The alternative to fired brick was an interlocking block that can be produced locally with the use of local materials, human resource and is durable. Build Up Nepal encouraged local entrepreneurs to involve in production of CSEB interlocking bricks for an alternative to fired brick. Such a brick would not need to be burned but rather be compressed in the machine and then cured with water for 21 days. It can be produced using local material like mud, sand and cement (10%). With the installation of a simple compressive machine which cost NPR 205,000, one can produce CSEB interlocking bricks with additional NPR 205,000 of material at community level for reconstruction at a low price. It has been used for more than 30-40 years all around the world in places like India, Thailand, New Zealand, USA and Malaysia. It is earthquake resistant as the bricks interlock to each other like big LEGO, and with cement & rebar reinforcement, the bricks are made even more disaster resistant. Using cement or rebar placed inside the holes of the brick makes the whole house act as a single unit and highly disaster resistant.

To effectively involve households in using and understanding the new technology, local entrepreneurs must be encouraged to start and learn this technology, learn some of the ways in which these bricks are being marketed and start selling using local resources. Acceptance of the community is the most important. This will make them more sustainable and cost effective on building new houses. The cost of construction is reduced by approximately 25%. It is cheaper than using mud mortar, stone masonry with cement bands because the interlocking bricks just need a small amount of cement to pour into the holes. Two roomed houses can be built in NPR 400,000 or less saving the cost of plaster. The thermal insulation

property that allows the houses made from these bricks to remain cool in the summer months and warm during the winter months, that has been very beneficial for the house owners and something that is being explored more for marketing purposes. We can compare on the material cost between fired bricks and CSEB interlocking blocks as follows:



1 m² Interlocking Brick (CSEB) wall

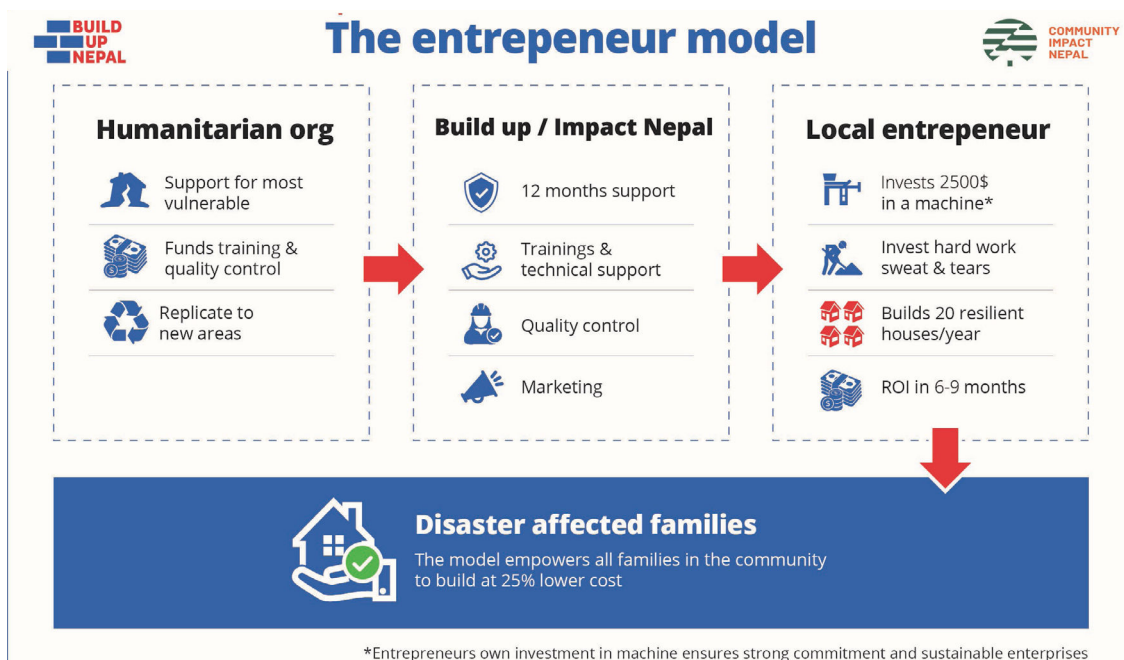
Material	Quantity	Rate	Cost
	32 pcs (30x15x10 cm)	50 rs	1600 rs
	1.5 hours	125 rs	188 rs
	8 kg	22 rs	176 rs
	1 cubic feet sand	100 rs	100 rs
		Total cost	2064 rs/m²



1 m² fired brick wall

Material	Quantity	Rate	Cost
	140 bricks (22x11x5.5 cm)	15 rs	2100 rs
	2.16 hours	125 rs	271 rs
	28 kg	22 rs	616 rs
	2.74 cubic feet sand	100 rs	274 rs
		Total cost	3261 rs/m²

Build Up Nepal & Community Impact Nepal has supported 295 enterprises & communities to build 4850 houses creating 2400 jobs, both in production and construction reducing 20,610 tons of CO₂ emissions. Support for different organizations are available if the government wants to promote this technology aiming towards a better community.



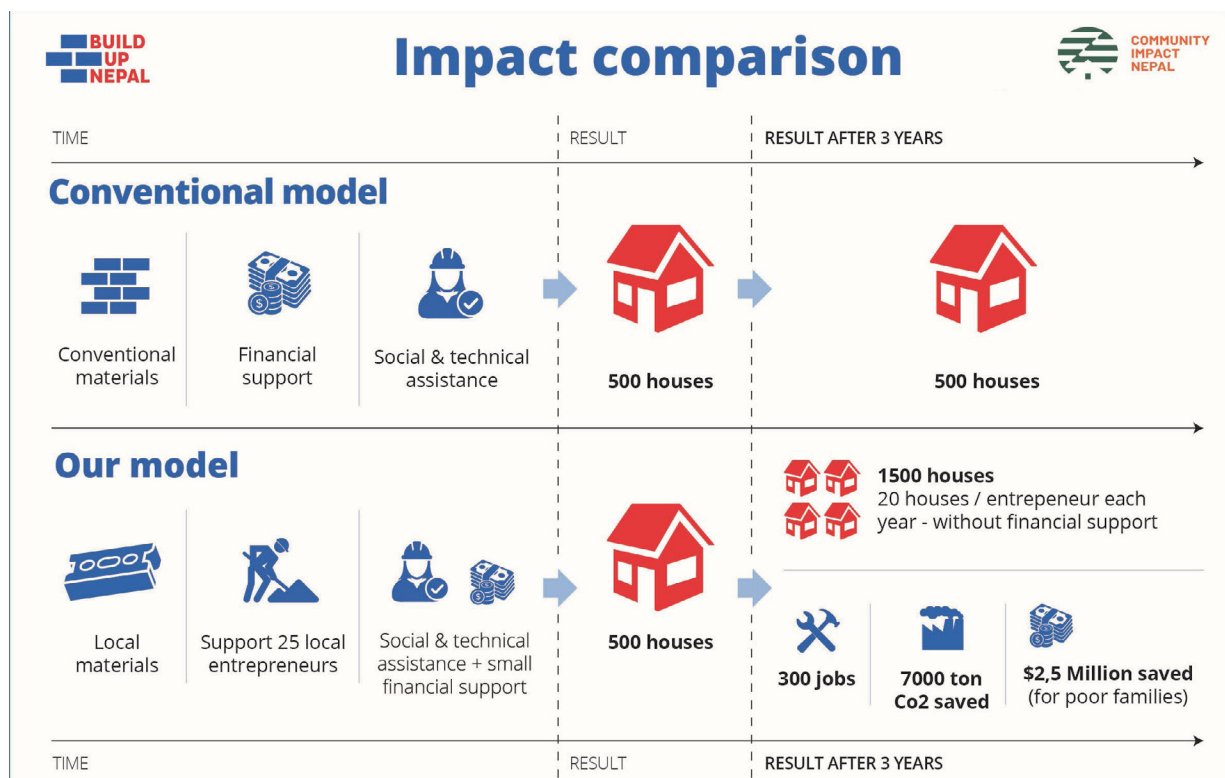
IMPACT COMPARISON

By combining the local entrepreneurship model and Compressed stabilized earth bricks, Build up Nepal and Community Impact Nepal have found a very innovative solution for affordable housing reaching higher number of users using less funding from the projects.

The idea:

Local resources cost less and combine with the local entrepreneur who has the know-how of the area and is very effective in getting the work done at their community level drives the cost further down. In the conventional housing projects, the funds are allocated for mainly socio-technical assistance, top up grant and use of conventional building material mainly fired bricks.

By using CSEB, we can reduce the building cost by 25% and that money can be used to support the local entrepreneurs who will continue to build even after the project is complete and the target household has been reached. By supporting the local entrepreneurs, we can attain the sustainability part by having a vision beyond the project. By establishing these small microenterprises, we will be able to continue building, create local jobs and save money for the community and the added benefit of course of saving the harmful CO2 emissions coming out from other building materials. Please see the picture.



Remarks from Dr. Chandra Bahadur Shrestha, Executive Committee Member, NRA

As NRA is at concluding mode, the experiences and practices that different partner organizations have made must be recorded, shared and handed over to the future generations around the globe. All of us are keen to learn what were the innovations, what were the challenges faced and how were they resolved? There are challenges but opportunities as well.

We have reached the point where quality control has become the major issue. And the experience gained out of Cement hollow blocks is unsatisfactory. A worry on the sustainability of CSEB entrepreneurs and promoters are also valid. The guideline produced by the partner organization will be reviewed by respective CLPIUs and a proper norms or guidelines will be established from the government which is very indispensable at local level. It is required to ensure that proper information is disseminated, increase the acceptability of this technology and a mechanism to ensure quality assurance at local level.

At the federal state, the District Level Implementation Unit (DLPIU), Building can be recommended to the CLPIU but it must be approved by the local government. If the proposals are supported by the facts and figures, the recommendations made by the central government are eventually endorsed by the local government. Local entrepreneurs should use their leverage to create an environment for effective consumption of CSEB interlocking blocks by various households.

NRA will be nurturing by engaging and coordinating with the CLPIUs and central government to resolve any difficulties of approvals in creating an environment where CSEB entrepreneurs and promoters are supported. Engagement of department of cottage and small industry (DCSI) for appropriate specification of CSEB interlocking block for the entrepreneurs is crucial at local level. DCSI can disseminate the information related to the CSEB interlocking block at community level which eventually can promote it. The established CSEB association should engage with the district level cottage and small industry so that it can be institutionalized. The cottage and small industry is then able to provide information, conducting promotional camps and technical training on it interlinking with other stakeholders like banks, DLPIU/Building, DLPIU/Education, etc. In the process, NRA can facilitate interlinking the logical sequences lubricating it for quick results.

In terms of Kathmandu Valley, if this can influence consumers at a higher scale the CO₂ emission out of fired bricks, larger scale of production has to be done. For this more work

and effort is still needed. The operation is at the local level at a small scale with limited numbers. If the compressive strength of CSEB is sufficient then it can be even used in RCC structures. The next step to contribute to control CO₂ emission can happen if it is produced and distributed at a large scale. An area of research could be to find out if it can be produced with the available soil quality at Kathmandu valley and other places of Nepal in order to produce CSEB bricks at various places. The payback period of CSEB of 6 - 9 months is also appealing.

A case of Rasuwa district where houses built with CSEB if exposed to rain up to the ceiling level cement plaster will be required. In such a situation stone dust was suggested to be used as an alternative to mud but there might be difficulty of transportation as well. Also, the essential raw materials for CSEB is sand and cement which might be difficult to transport at remote areas and therefore, such limitations must be mitigated for the support of vulnerable households. Beside this, the equipment used for the production of CSEB should also be manufactured locally so that establishment of new enterprises can be easy. As the impact of earthquake reconstruction houses is approaching at closer/end the question might be at replicability of this technology in future.

Mr. Deepak Hamal, President of Interlocking Bricks Association of Nepal (IBN)
Proprietor of Easy Interlocking Bricks, Palung, Makwanpur district, Nepal

Interlocking Bricks (also known as CSEB – Compressed Stabilized Earth Bricks) are made locally, reducing cost, and making the dream of a Brick house affordable. Interlocking Bricks (CSEB) are made by mixing and compressing Sand, Soil and Cement in a machine. It can also be made from stone-dust and cement. The bricks are then stacked and cured for 21 days – the cement sets and bonds with the sand stabilizing the brick.

When building a house, we utilize the Interlocking Brick Masonry and reinforce the walls with rebar, both horizontally and vertically, adding to the strength and making the structure earthquake resistant.

Easy interlocking bricks have started its operation from April 2017 (Chaitra 2073) by deploying 15 local field staff. Build Up Nepal provided technical support on interlocking bricks and started operation thereafter. Since then earthquake housing reconstruction has been given first priority.

Leave No One Behind (LNOB) related projects supported by Build Up Nepal and Practical

Action Nepal have initiated to support more than 400 houses so far with interlocking bricks.

Community people with low economic category used to build 2 roomed houses by using local red bricks amounting to a total of 4-5 lakhs. However, after intervention of LNOB projects, now earthquake affected beneficiaries have rebuilt houses amounting 3.5 to 4 lakh houses per house by using interlocking bricks. LNOB projects provide additional NPR 50,000 for vulnerable beneficiaries (and those who cannot afford) that are using interlocking bricks.

Vertical rebar are put in each corner, at doors and windows and evenly spaced throughout the walls. This is then connected with horizontal seismic bands making the whole structure interconnected and earthquake resistant. The government of Nepal approved the Interlocking Brick technology in 2017 in earthquake housing reconstruction.

Interlocking bricks industry has many benefits as the raw materials are locally found and it generates employment for local people. The quality control is the challenging factor at operational level as its quality has to be tested and maintained for uniformity and consistency while the lack of clarity on interlocking bricks and no guidelines/norms that defines use of interlocking bricks for other normal constructions also is needed at the moment.

**Mr. Dadhiraj Pudasaini, Proprietor of Indrasarowar Itta Udhyog,
Palung, Makwanpur district, Nepal**

For sustainability purposes, quality control has to be taken seriously. The interlocking bricks that are being produced by the entrepreneurs supported by Build Up Nepal and Community Impact Nepal, are producing with quality but there are others who have bought the machine from India and elsewhere from Nepal, whose quality is not uniform and needs to be monitored. In the beginning, two of my friends and I decided to invest in the machines and start producing these bricks, we invested in 3 machines and started the factory from the same location. There were many challenges ahead of us as we were confused for upto 6 months thereafter we approached local governments in our district, initiated interaction programs, made demo structure and conducted dissemination which later led to proper operation. Insights were provided by engineers of Community Impact Nepal, a local NGO in taking interlocking bricks in earthquake housing reconstruction with sustainable measures ahead. Later we were also approached by NABIN Project (supported by Community Impact Nepal, Practical Action and Dan Church Aid) that led to conducting interactions at community level by involving local stakeholders and community people. Slowly, local

stakeholders were convinced with our work and were valued among earthquake beneficiaries thereafter. In our area and in surroundings too, government buildings mainly schools are also using interlocking bricks in construction and other some non-earthquake beneficiaries too are using interlocking bricks in construction. So, in general, the support that we received from these organizations and projects helped us to convince a lot of important stakeholders who later became our customers. Even the locals started trusting the bricks and the technology once it was endorsed from the local government. Gradually, even without the support from the projects and these organizations, our market has grown exponentially with a little hiccup during the COVID time.

Quality of interlocking bricks produced throughout the country must be stressed for creating uniformity. Our association has to be capacitated and the funds have to be managed if we want the sustainability for this technology. We, as entrepreneurs believe that this is the best possible answer to replace the fired bricks not just in terms of affordability but also carbon friendly and create local jobs but our commitment has to be backed by the government and the central bodies.

ANNEX A: QUESTIONS AND ANSWERS (Q&A)

Q: What specific supports need to be done to promote interlocking bricks?

A:

There is a need for a quality test done of interlocking bricks once in every 3 weeks, so that quality is assured throughout and monitored too. Government proper guidelines on inclusion of interlocking bricks as integral part like general red bricks should be carried forward. Minimum quality tests of interlocking bricks should come upto 5-10 mpa.

Interlocking bricks have been discovered to use locally available sand and soil & made by local labourers too so its usage and making is environmentally friendly.

Q: What kind of support and cooperation is expected from the Government & POs?

A:

The current NABIN project may phase out in near future and its sustainability is challenged thereafter. Therefore, to sustain interlocking bricks in longer runs, the government should own it by endorsing and promoting through guidelines that should address quality control primarily. There are challenges of ceasing this promising industry once earthquake housing reconstruction stops as these bricks are mostly promoted and utilized in earthquake housing reconstruction. Interlocking Brick Association Nepal (IBN) firmly stands with a position that guidelines could help in promoting its usage by other general people in other public and private construction.

Q: How much demand is perceived from non-earthquake affected people regarding Interlocking bricks? How much demand is created from the government side? How does the community take the interlocking bricks?

A:

Currently interlocking has been used by earthquake housing reconstruction and as long as NABIN Project remains but once it is phased out there are challenges in its future sustainability. These guidelines from the government in its inclusion, quality control and promotion. Also there is a need for district rate maintained by the government should promote the construction of public buildings like schools and other community infrastructure. Once guidelines are on board, then IBN is ready to take responsibility for assurance on quality and monitoring too.

Q: How IBN cooperates with the interested person to invest in interlocking bricks?**A:**

The other people who want to initiate this entrepreneurship should follow the general rules regulations set by IBN and should keep a transparency board that mentions them. This material should not be taken as a block, it should be kept in a category of bricks that uses local raw materials and made from local labours.

Q: What is best for Earthquake prone areas especially in Mountains?**A:**

CSEB has been very popular in the Hilly region of Nepal. Especially where people aspire to build brick houses but poor roads make transportation makes brick very expensive.

For construction above 2500 meter, we recommend avoiding brick production during December - February as the bricks will not cure properly in sub zero temperatures.

Q: Is CSEB equally good to use in houses of Terai region in Nepal where it is intense heat and rain?**A:**

One of the properties of CSEB bricks is the thermal insulation that allows the temperature inside the house remain unaffected regardless of the temperature outside. In case of extreme heat, the CSEB house owners in terai have complimented that the house remains cool inside and vice versa during the winter months.

During the rainy seasons, the problem of seepage was one of the biggest worries in the beginning but after we practiced putting the 5 mm mortar in brick laying, that problem was addressed and corrected. The houses made after August, 2017 and have seen three rainy seasons have had no problems with the rain. Of course, for flooding, we need to raise the plinth to make sure that the overall height of the house itself is lifted. Also in flooded areas we recommend making CSEB from Stone Dust and cement. Such bricks are very water resistant and do not lose more than 10-15% compressive strength when submerged in water.

Q: How does the interlocking bricks contribute to resilient livelihoods? How does it benefit the vulnerable people?**A:**

We have a combination of CSEB technology and local entrepreneurship through which we have been able to contribute to resilient livelihood. We have so far reached more than 250

communities, meaning one factory to produce these bricks in each of these locations and helping create at least 5 jobs in production and 5 in construction.

CSEB bricks can lower the cost of construction of a house by at least 25% and in some cases upto 40%. Actually the highest uptake of the technology is among poor families due to the high cost savings. Out of 4500 houses built, more than 1500 houses are for highly vulnerable families through various projects.

Q: How did you manage the sites from where the earth was excavated from the Natural Resource Management lense? How feasible was it to manage transportation of CSEB in the Mountain terrain from the production site?

A:

Our production is small scale and topsoil can not be used to make bricks. Most entrepreneurs take soil either from road construction or by digging out a terrace and keeping the topsoil intact. In some locations our entrepreneurs use stone dust as raw materials. Stone dust CSEBs are very strong and solves our whole problem with natural resources management. To be fair, by using stone dust, we have been able to reduce the cost of the bricks and able to use something that was polluting the river beds. The stone dust was anyways being dumped into the river along the highways, now that they are being used to make the bricks.

In very remote locations without road access it is often effective to transport the machine instead of the bricks. Our small machine can be dismantled in parts and moved between construction sites.

Q: I would like to learn about the production and testing of the CSEB, Engineering properties, the blocks, perception of house owners, the price of machines etc.

A:

The bricks are made from sand, soil and 10% cement. The composition of soil/sand is very important and our training focuses much on soil testing to get the correct mix proportions. House Owners like the bricks because they want to live in a brick house and CSEB makes it affordable. They also like to see small concrete columns and rebars going through the bricks as they can easily understand how it is earthquake resistant.

Machines cost 235,000 to 365,000 rs for manual versions. There are also hydraulic machines available which are more expensive and efficient.

Q: How about housing insurance for the poor people?

A:

We don't have any program for housing insurance currently.

Q: What is the accessibility of CSEB to architects, artisans and local people?

A:

From our organisational perspective, we would like to have as many people interested to learn about CSEB as possible. In fact, since we are also learning with each passing day, we encourage the architects, engineers and artisans to challenge us with more questions so we can find appropriate answers for them. We have had architects doing their project with us for different universities in India, Europe and Australia.

Q: How much is needed for a local entrepreneur to start an interlocking brick business? Is there a specific soil needed for this technology?

A:

A local entrepreneur will have to invest anywhere between NRs 400,000 to 800,000 depending on the machine and the factory set up costs and overheads. (more in case of hydraulic machines and large scale production).

The training is paid from the project, lowering the investment for the entrepreneurs and making it easy for them to return their investment within a year.

Q: I would like to know more about employment creation at local level from this project and how GESI has been incorporated.

A:

The enterprises hire local people to make bricks creating local employment. We have found that brick making is a good job for women and have on average 30-40% women in the production in most enterprises.

Q: How does the local availability of sand impact the feasibility and sustainability of the technology?

A:

For areas with less sand availability, CSEBs can be made using 10 % cement and 90% stone dust. We had this issue in some of the sites especially in Gorkha where we experimented with the stone dust and the bricks were even stronger. But either sand or stone-dust is required to produce good quality CSEB unless you have highly sandy soil.

Q: Introduction of either new technology or construction materials was the missed opportunity in Nepal's reconstruction. What were the key constraints for this?

A:

Technical problem and sustainability

To ensure sustainability, Build up Nepal and Community Impact Nepal will continue to develop the technology, to enhance its efficiency, provide need based technical support and improvised the business model for each enterprise. Apart from this, there is a national level association registered and both the organisations are looking to work with them in the future to make sure the gaps in the construction and production are addressed even after the project.

Q: What are other Alternate low cost housing material, design and methodology beside CSEB block method?

A:

Rammed earth, Stone masonry, GI wire, Bamboo houses are just some of the low cost housing solutions but CSEB technology has been found to be the most economic and socially accepted.

Q: Share the NABIN project updates?

A:

Objectives of the NABIN project:

- Creating an enabling business environment for climate friendly construction material*
- Reducing the impact of climate change through mitigation and partly, through adaptation and resilience*
- Job creation and livelihood promotion in project areas and support to poor and vulnerable groups, and*
- Climate change advocacy and policy influence to secure a low emission path way for the Nepalese public and private sector.*

(The project will contribute to SDGs 9 and 13)

Closing Remarks

Mr. **Shyam Kishor Singh**, Chief, NRA CLPIU Building

Shelter construction for the poor and vulnerable is always a priority task for the government. NRA since its earlier phases have prioritized the rehabilitation of vulnerable and poorest of the poor earthquake affected people by providing additional Rs 50 thousand along with

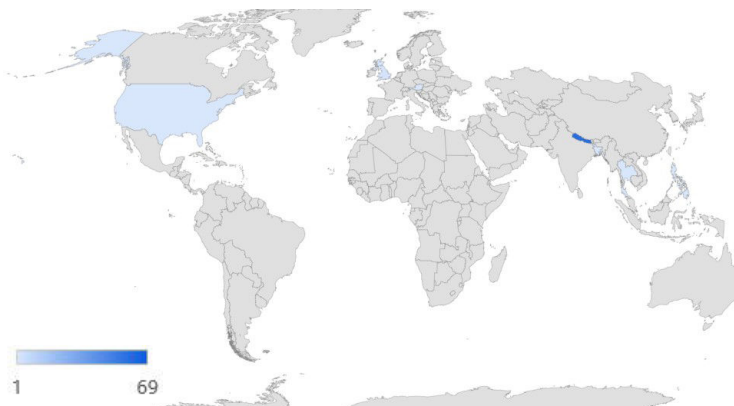
300,000 NPR tranche amounts. There are nearly 18,500 vulnerable beneficiaries for whom houses need to be reconstructed. NRA's decisions to mobilise mobile masons and social mobilizers from last fiscal year in reconstruction of houses for vulnerable beneficiaries were hampered by the current COVID 19 Crisis that not only disturbed the related supply chain system but also restricted the movement of those mobile masons and social mobilisers to unreached people. Despite this, our decisions to support and work for vulnerable people are intact and will continue fulfilling our commitments. Studies revealed that financial constraints are the biggest challenges for vulnerable beneficiaries to initiate reconstruction as around 5-6 lakh rupees is required in whole construction begging from naksha making (building drawing) to completion. Even though NRA provides Rs 350000, there have been cases revealed that those identified vulnerable cannot afford the rest of the extra amounts. But with interlocking bricks, it was found that houses for vulnerable and poor people can be made by an amount ranging from Rs 350000 to 400000/-

Key suggestions

- To internalize this interlocking bricks technology, there is need of massive awareness at local level, should be disseminated among communities for convincing to best, should be mobilized and owned by local governments as well.
- The key messages of environmental friendly benefits and local employment creation should be imparted among public for its larger acceptance in longer run so that other interested business minded people could also start this technology by investing in a machine worth Rs 25000 only.
- Local governments should be approached for the usage of interlocking bricks in their public infrastructure construction program so that ownership is generated widely.
- DUDBC as a key line agency in building construction should be approached with detailed technical and financial proposals by IBN which will be sent further to the central government.
- For district rate matters, IBN should emphasize District Administration Office through regular coordination and once approved by DAO, local government units (Gaunpalika and Nagarpalika) will own it by endorsing- this is the second way forward.
- Usage of interlocking bricks should be massively promoted and this business should have an ability of demand and supply from other districts and by non-earthquake beneficiaries too. Only then this reflects the multiplier effect in rampant usage.
- For assuring quality control, the raw materials used should have soil and lab tested, the report of which assures the quality among producers.

ANNEX B: PARTICIPATION DETAILS

Country	Registered	Attended
Austria	1	0
Bangladesh	1	0
Nepal	60	63
Philippines	1	1
Singapore	2	0
Thailand	1	0
United Kingdom	2	1
United States	1	0
Total	69	65



Geographic distribution of the webinar participants

SN	Name	Affiliation	Country
1	Aanand Prasad Gupta	National Reconstruction Authority	Nepal
2	Abhishek Shrestha	Habitat for Humanity Nepal	Nepal
3	Ajay Pandey	DCA	Nepal
4	Ananda Manandhar	Lumanti Support Group for Shelter	Nepal
5	Andreas K	Build up Nepal Engineering	Nepal
6	Arjun Sharma	Infrastructure Development Office, Myagdi	Nepal
7	Ashish Rai	National Reconstruction Authority	Nepal
8	Asmita Shrestha	Lumanti support group for shelter	Nepal
9	Babita Sharma	NA	Nepal
10	Bhubaneswari Parajuli	HRRP	Nepal
11	Björn Söderberg	Build up Nepal Engineering	Nepal
12	Caroline Wu	Operation Hope Foundation	Singapore
13	Cecilia	IFRC	Austria
14	Chandra Hada	DAI	Nepal
15	Charles Parrack	Oxford Brookes University	United Kingdom
16	Danu Darai	HRRP	Nepal
17	Deepak Dhungel	AIDS Healthcare Foundation (AHF) Nepal	Nepal
18	Deepak Saud	HRRP	Nepal

SN	Name	Affiliation	Country
19	Dev Raj Joshi	Dogadakedar gaupalika	Nepal
20	Durga Prasad Dhakal	ChildFund Japan	Nepal
21	Dwarika Shrestha	Trust Consulting	Nepal
22	Er. Bijay Raj Neupane	DLPIU, Chitwan, Bharatpur Metropolitan City	Nepal
23	Hari Bahadur Karki	United Nations Population Fund	Nepal
24	Hom Pathak	Nepal Redcross Society	Nepal
25	James T Kendall	Habitat for Humanity	Philippines
26	Jamie Richardson	CRS	United Kingdom
27	Karishma Thakur	USAID Tayar Nepal	Nepal
28	Kausal Bist	HRRP	Nepal
29	Krishna Prasad Osti	ASHA Project, Ministry of Forest and Environment, Nepal	Nepal
30	Lumanti Joshi	Lumanti Support Group for Shelter	Nepal
31	Madhukar Shrestha	Nepal Red Cross Society	Nepal
32	Mahendra Shakya	Lumanti Support Group for Shelter	Nepal
33	Mehul	Catholic Relief Services	Bangladesh
34	Merina Bajracharya	UNOPS	Nepal
35	Mim Hamal	EU	Nepal
36	Krishna Karkee	USAID/Tayar Nepal-Improved Disaster Risk Management	Nepal
37	Nishit Mittal	National Structure & Engineering Pvt Ltd	Nepal
38	Parikshya Singh	Lumanti Support Group for Shelter	Nepal
39	Pooja Shrestha	VSO	Nepal
40	Prabin Gautam	HRRP/CRS	Nepal
41	Prakash Timalsina	United Mission to Nepal	Nepal
42	Pratik Singh Parmar	Build up Nepal/ Community Impact Nepal	Nepal
43	Rabindra Das	Transform Nepal	Nepal
44	Radhe Shyam Rai	Organizational Development Support	Nepal
45	Rajani Thapa Magar	Practical Action, Nepal	Nepal
46	Rajesh Rai	People in Need	Nepal

SN	Name	Affiliation	Country
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48	Raju Chhetry	UN	Nepal
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53	Rudra Neupane	PHASE Nepal	Nepal
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60	Samjhana Rajbhandari	National Reconstruction Authority	Nepal
61	Shankar Pokhrel	NRA - DLPIU Building Kathmandu	Nepal
62	Shubhadra Devkota	Danish Red Cross	Nepal
63	Shuva Sharma	Scott Wilson Nepal	Nepal
64	Sunita Kutu	National Reconstruction Authority	Nepal
65	Sushil Ojha	Shinawatra University	Thailand
66	Udaya Bikram Thapa	World Vision International Nepal	Nepal
67	Vishma Basnet	NSET-HRRP	Nepal
68	Wesley Goh	Operation Hope Foundation	Singapore
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